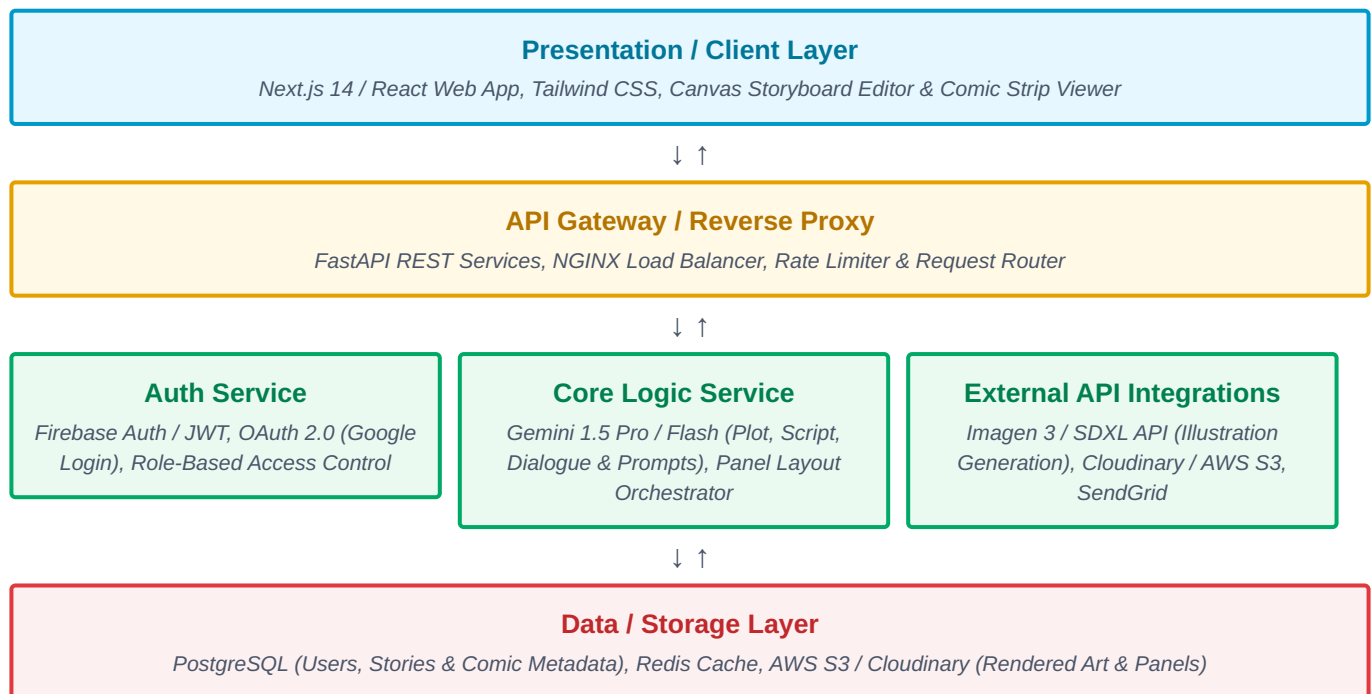


Solution Architecture

Date	15 March 2026
Team ID	UNM293SmartStudyAI
Project Name	ComicCraft - AI Comic Story Creator using Gemini Models
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation / Client Layer	Provides a responsive, interactive user interface allowing creators to input story premises, character definitions, genre preferences, and comic panel configurations. Features real-time comic strip rendering, storyboard drag-and-drop sequencing, and PDF/image export preview.	Next.js 14 React Tailwind CSS HTML5 Canvas TypeScript
API Gateway & Reverse Proxy	Handles all incoming HTTP/HTTPS requests from the front-end, enforcing TLS termination, IP-based rate limiting, load balancing, request validation, and routing requests to appropriate backend services.	FastAPI NGINX Uvicorn Docker
Authentication Service	Manages user registration, session management, secure credential handling, and profile persistence. Grants authenticated tokens to protect user drafts, comic libraries, and AI credit balances.	Firebase Auth OAuth 2.0 (Google) JWT Tokens Bcrypt
Core Logic Service (AI Story Engine)	Executes prompt engineering pipelines using Gemini models to generate multi-panel narrative arcs, character dialogues, speech bubbles, sound effects (SFX), and detailed visual panel prompts for artwork generation. Orchestrates comic layout stitching.	Google Gemini 1.5 Pro Gemini Flash Python 3.11 LangChain Pillow (PIL)
External API Integrations	Connects to generative image generation backends to render illustrated panels based on Gemini-generated scene descriptions. Also handles transactional emails for comic exports and project sharing.	Google Imagen 3 API Stable Diffusion XL SendGrid API Cloudinary SDK
Data & Storage Layer	Stores relational structured data including user accounts, comic metadata, panel sequences, prompt history, and user settings. Manages cloud blob storage for high-resolution panel illustrations, final compiled comics, and caching for fast panel retrieval.	PostgreSQL Redis Cache Amazon S3 / Supabase SQLAlchemy ORM